LINEAR REGRESSION

Understanding how machines learn relationships

and predict numerical values

$$\hat{y} = b_0 + b_1 x$$



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What Is Linear Regression?

Linear Regression is a supervised machine learning algorithm used to predict a numerical value from one or more input features.

PREDICT A PRICE

Estimate a house price from its area.

PREDICT A SCORE

Estimate an exam score from study hours.

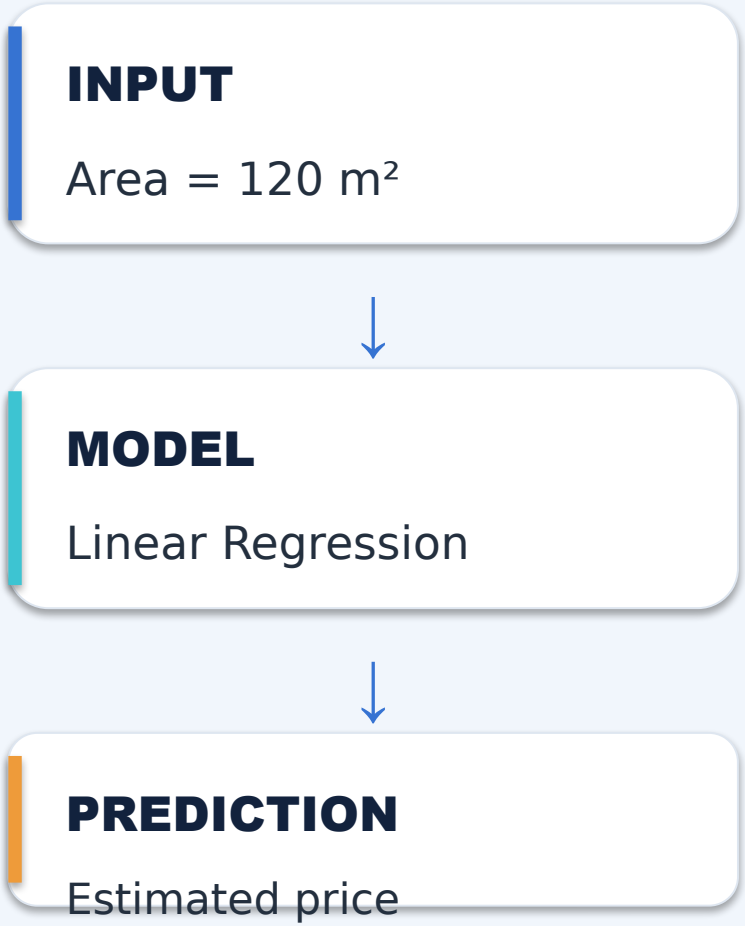
PREDICT A VALUE

Estimate a used car's selling price.

Key idea: learn a relationship from historical data, then use it to make predictions.

When Do We Use It?

- House prices based on area, location, and other features.
- Exam scores based on study hours.
- Used-car prices based on age, mileage, and model.
- Sales or demand forecasts based on historical measurements.



A Simple Example: Study Hours and Scores

Suppose we collect these training examples:

Study Hours (X)	Exam Score (Y)
1	50
2	60
3	70
4	80

OBSERVATION

Each extra study hour is associated with 10 more points in this small example.

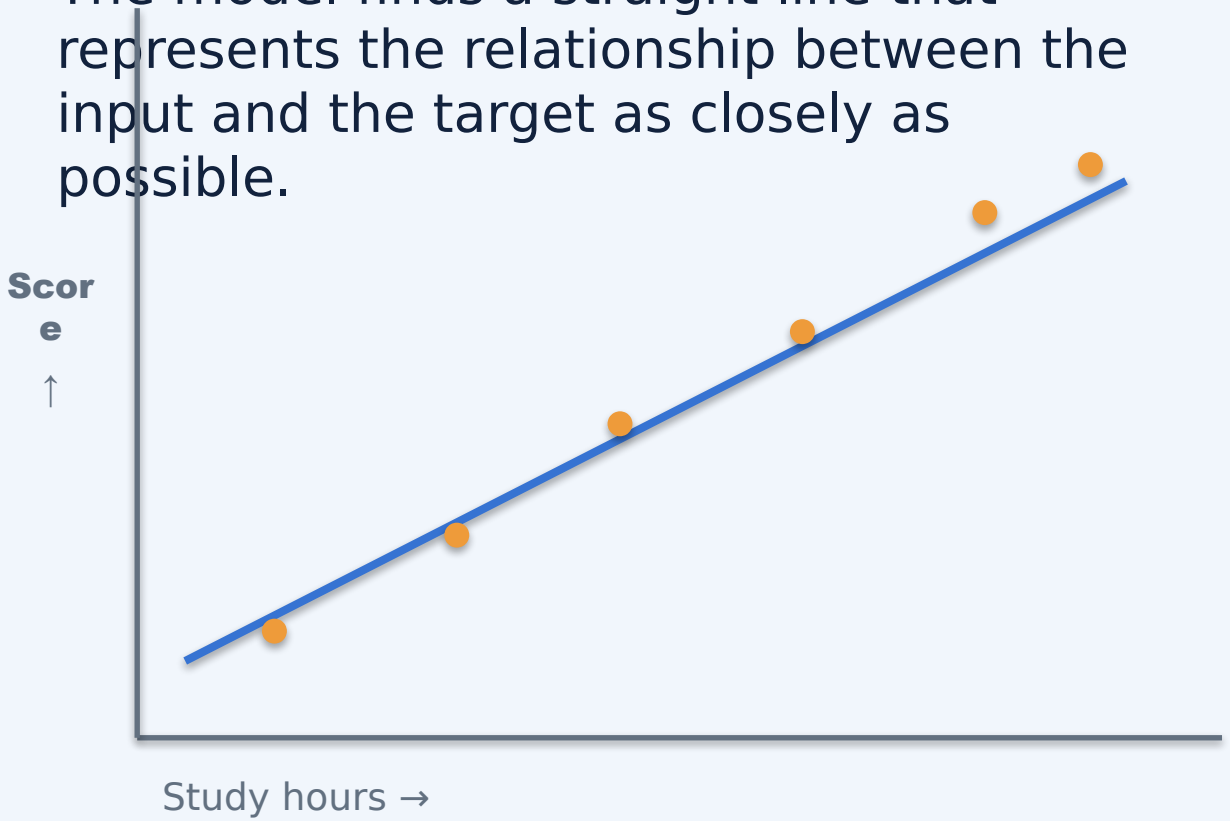
PATTERN

The data follow a straight-line relationship.

This pattern is illustrative—not a universal rule about studying.

The Best-Fit Line

The model finds a straight line that represents the relationship between the input and the target as closely as possible.



OBSERVED DATA

Dots represent the actual values in the dataset.

BEST-FIT LINE

The line captures the overall trend and helps predict new values.

Explore the relationship by moving points in the interactive regression demo (if available).

The Linear Regression Equation

$$\hat{y} = b_0 + b_1 x$$

$\hat{y}$ — PREDICTION

The numerical value estimated by the model.

x — INPUT

The feature used to make a prediction.

b_0 — INTERCEPT

Predicted value when x = 0.

b_1 — SLOPE

Expected change in $\hat{y}$ for a one-unit increase in x.

The values of b_0 and b_1 are learned from the training data.

Using the Equation

For our study-hours example:

$$\hat{y} = 40 + 10x$$

If a student studies for 3 hours:

$$\hat{y} = 40 + 10(3) = 70$$

INTERCEPT = 40

The line starts at 40 when x = 0.

SLOPE = 10

Each additional hour increases the prediction by 10 points in this example.

For 5 study hours, the model predicts 40 + 10 × 5 = 90.

How Does the Model Learn?

1. Collect data

Gather input features (X) and actual target values (y).

2. Fit the model

Estimate the intercept and slope from the training examples.

3. Calculate errors

Compare each prediction with its actual value.

4. Minimize error

Ordinary Least Squares minimizes the sum of squared residuals.

Residuals: Measuring Prediction Error

A residual is the difference between an actual value and its prediction.

Residual = y − ŷ

EXAMPLE

Actual score = 75

Predicted score = 70

Residual = 75 − 70 = 5

WHY SQUARE THE ERRORS?

- Positive and negative errors do not cancel.
- Large errors receive more penalty.
- The model can compare candidate lines consistently.

Ordinary Least Squares chooses coefficients that minimize the sum of squared residuals.

How Do We Evaluate the Model?

MAE — MEAN ABSOLUTE ERROR

Average absolute difference between actual and predicted values.

MSE — MEAN SQUARED ERROR

Average of squared prediction errors.

RMSE — ROOT MEAN SQUARED ERROR

Square root of MSE; expressed in the target's units.

R^2 — COEFFICIENT OF DETERMINATION

Measures performance relative to predicting the target mean.

Lower MAE, MSE, and RMSE are generally better; higher R^2 is generally better. R^2 can be negative.

A Simple Python Example

```
from sklearn.linear_model import LinearRegression

X = [[1], [2], [3], [4]]
y = [50, 60, 70, 80]

model = LinearRegression()
model.fit(X, y)

prediction = model.predict([[5]])
print(prediction)  # [90.]
```

LinearRegression()

Creates the model.

fit(X, y)

Trains the model.

predict([[5]])

Predicts the score for 5 hours.

X is 2D: rows = samples, columns = features.

Linear Regression vs. Logistic Regression

LINEAR REGRESSION

- Predicts a numerical value.
- Example: house price or exam score.
- Output is a continuous quantity.

LOGISTIC REGRESSION

- Commonly used for classification.
- Example: churn or no churn.
- Can estimate class probabilities.

For a telecom churn project, Logistic Regression may predict whether a customer leaves (1) or stays (0).

Strengths and Limitations

STRENGTHS

- Simple and easy to interpret.
- Fast to train on many datasets.
- Provides a useful baseline model.

LIMITATIONS

- Assumes a linear relationship in its basic form.
- Sensitive to outliers.
- Can underfit complex patterns.
- Correlation does not prove causation.

Always check the data, assumptions, and performance on unseen test data.

Key Takeaways

- Linear Regression predicts numerical targets.
- It learns coefficients for a best-fit line from training data.
- Residuals describe the gap between actual and predicted values.
- MAE, MSE, RMSE, and R^2 help evaluate performance.
- Scikit-learn makes it straightforward to train and use the model.

Thank you! • Questions?

Quick Knowledge Check

A model predicts house prices using house area. What type of problem is this?

A. Classification

Predicting a category.

B. Regression

Predicting a numerical value.

C. Clustering

Grouping similar data points.

Correct answer: B — Regression, because the target (price) is numeric.